

md:complex^{GenX}

Exosomes & Polynucleotides^{Regen}

Cell^{Switch} technology for signalling skin regeneration and rejuvenation processes.



Regenerative medicine

The growing interest in maintaining a youthful and rested appearance has prompted an accelerated development of non-invasive and minimally invasive aesthetic approaches for skin rejuvenation and regeneration.

The population requiring mini-invasive aesthetic treatments is continuously expanding. The term “pre-juvenation” refers to a preventive approach of anti-aging procedures.

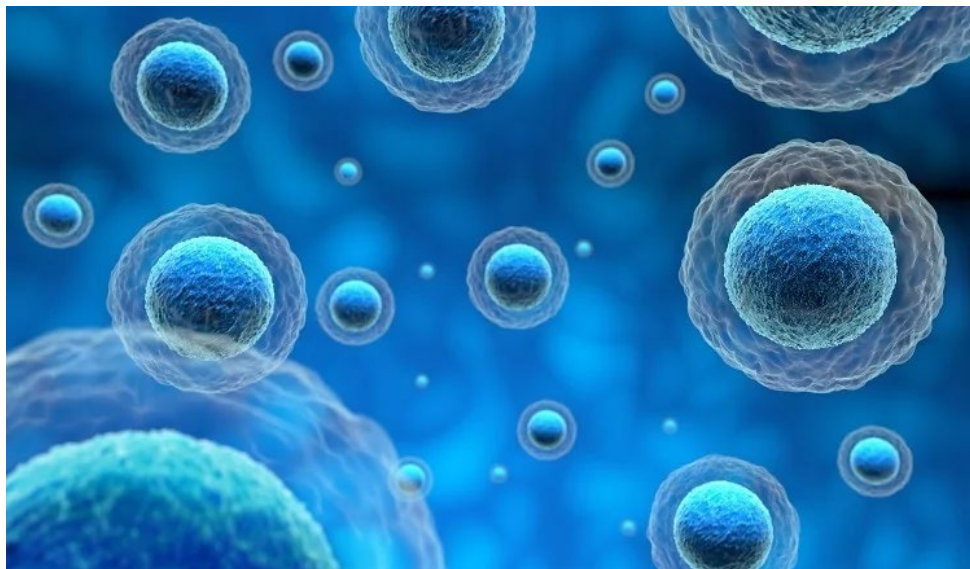
This innovative field sprouts regenerative dermatology as a combination of the principles of regenerative medicine in the context of skin regeneration and rejuvenation.



Regenerative medicine aims to replace lost or damaged tissues, while regenerative aesthetics focuses on regenerating soft tissues lost or damaged due to aging processes.

Regenerative medicine is focused on the repair, replacement or regeneration of cells, tissues, or organs to restore impaired function resulting from any cause, including ageing. The goal of regenerative medicine is to roll back the effects of age by utilizing the body's natural healing and regenerating capabilities.

In contrast to the traditional approach of primarily treating the skin ageing symptoms, the regenerative medicine focuses on tackling the concern from the source.

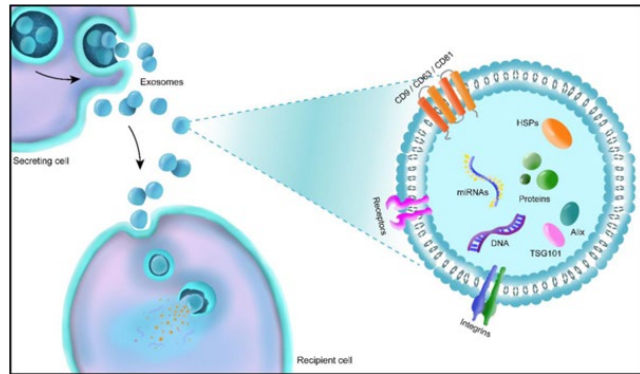


As the body ages, intracellular communication, and the body's powers of regeneration decline, which is why the effects of ageing become more pronounced.

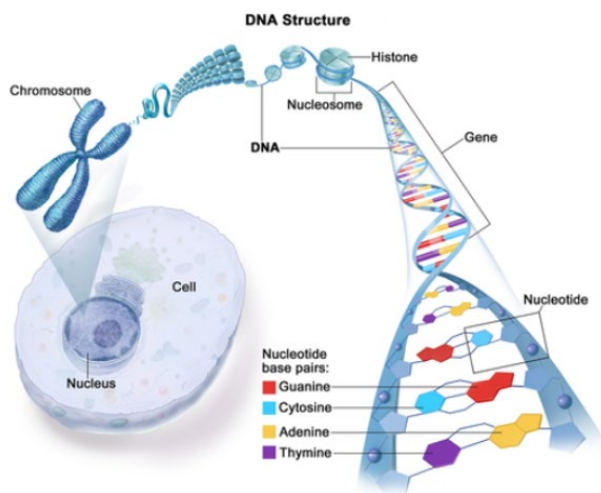
Regenerative medicine can use the power of cellular intelligence to reverse the ageing of tissues and revert the intracellular matrix back to a younger state. The discovery of the power of exosomes and polynucleotides is making a vital contribution to this field.

Exosomes are extracellular nano-sized vesicles secreted by cells, that transport essential cargo of lipids, proteins, nucleic acids (RNAs, mRNA, DNA), and metabolites required for cellular physiology as well as signalling molecules that facilitate intercellular communication.

They function as messengers for tissue repair, immune system modulation, and cell-to-cell communication.



Exosomes are released by donor cells to modulate the function of recipient cells. They can transfer essential information not only to neighbouring cells but also to remote cells, starting healing, regenerating, and rejuvenating processes.



Polynucleotides are long chains of nucleotides that make up DNA and RNA molecules, that function as a biostimulator, kick-starting regeneration in the skin.

They are key molecules for the vitality of the cells, providing sources of energy that drive most of the reactions.

Polynucleotides are a newly emerging trend in anti-ageing injections that stimulate tissue repair and encourage restoration and regeneration of the skin from within. They can be used in a wide range of applications as a rejuvenating treatment for older skin and as a preventative treatment for younger skin.

Properties:

md:complex^{GenX} Exosomes & Polynucleotides^{Regen} formulation combines the cell-signalling power of exosomes, polynucleotides and growth factors with the firming and hydrating effect of hyaluronic acid, which all work together to stimulate tissue repair and encourage restoration and regeneration of the skin from within.

md:complex^{GenX} Exosomes & Polynucleotides^{Regen} boosts natural cellular repair, enhancing the skin's own repairing properties.

It works by signalling to cells to change their behaviour and start processes of regeneration, repair, and collagen production, preventing chronological ageing. It improves elasticity and hydration, calms inflammation and rebalances melanocyte activity to create an even and refreshed skin tone.

md:complex^{GenX} Exosomes & Polynucleotides^{Regen} stimulates cell proliferation, improving the vitality of fibroblasts and the rebuilding of the connective tissues, leading to a more elastic and hydrated skin with a thicker dermis and a significant decrease of wrinkles. It allows a better tissue vascularization and cell nutrition enhancing the proliferation of endothelial cells and the increase of cutaneous brightness and glow.

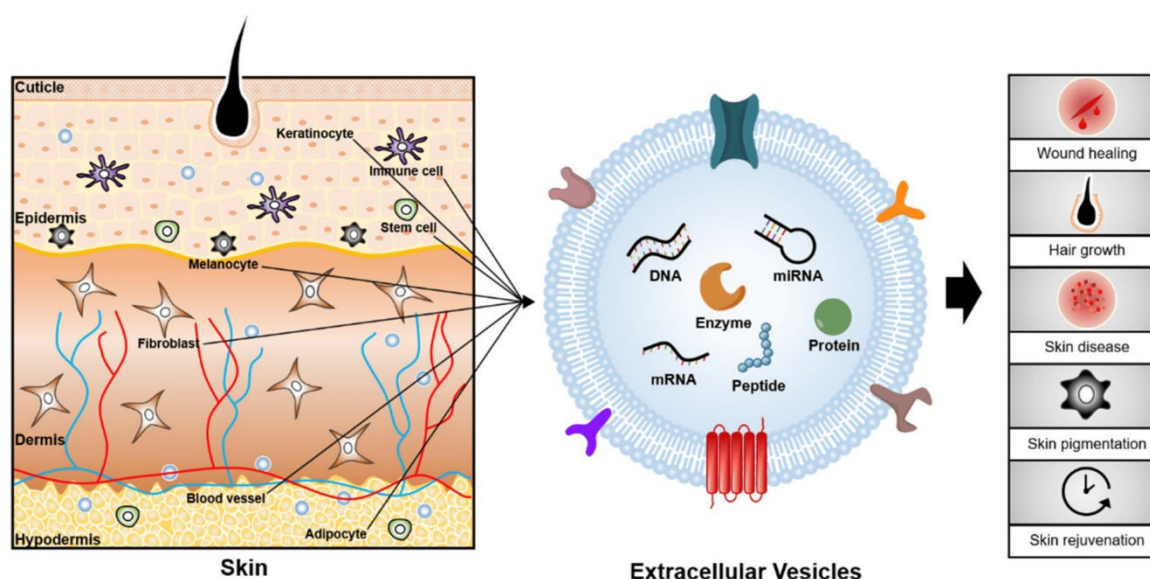


With high concentration of stem cells and growth factors, **md:complex^{GenX} Exosomes & Polynucleotides^{Regen}** restores the content of moisturising agents and lipids in the epidermis and strengthens the skin barrier against external aggressions. This allows the skin to regain volume, becoming plumped and radiant.

Benefits:

md:complex^{GenX} Exosomes & Polynucleotides^{Regen}:

- Improves cell-to-cell communication, carrying on essential information from donor cells to recipient cells to kickstart repairing and rejuvenating processes, improving cells' health. They transmit regenerative signals deep within skin cells, activating renewal processes.
- Enhances collagen and elastin regeneration and reverses fibroblast senescence, leading to the improved appearance of fine lines and wrinkles. Promotes the repair of photodamaged skin, helping to maintain skin elasticity and firmness.
- Accelerates wound healing, repairing skin damage caused by environmental factors or the natural aging process.
- Modulates the immune system to reduce inflammation and improve skin health. Regulates oxidative stress, protecting the skin from premature skin ageing. Calms irritated skin, reducing redness.
- Plumps and hydrates the skin, improving its texture. It attracts and binds water, helping to increase skin hydration and maintain its volume and turgidity.
- Stimulates neo angiogenesis and blood flow, ensuring optimal oxygenation and nutrition of tissues and increasing cutaneous brightness and glow.
- Blocks melanin production, leading to an even tone. Protects the skin from environmental stressors and reduces the appearance of dark spots and other discoloration.



Composition:

Centella Asiatica Extract Exosomes

Sodium Dna

Hydrolized DNA & DNA

Sodium Hyaluronate1

Rosa Centifolia Leaf Cell Extract

Centaurea Cyanus Callus Lysate Extract

Panthenol

COMPOSITION:

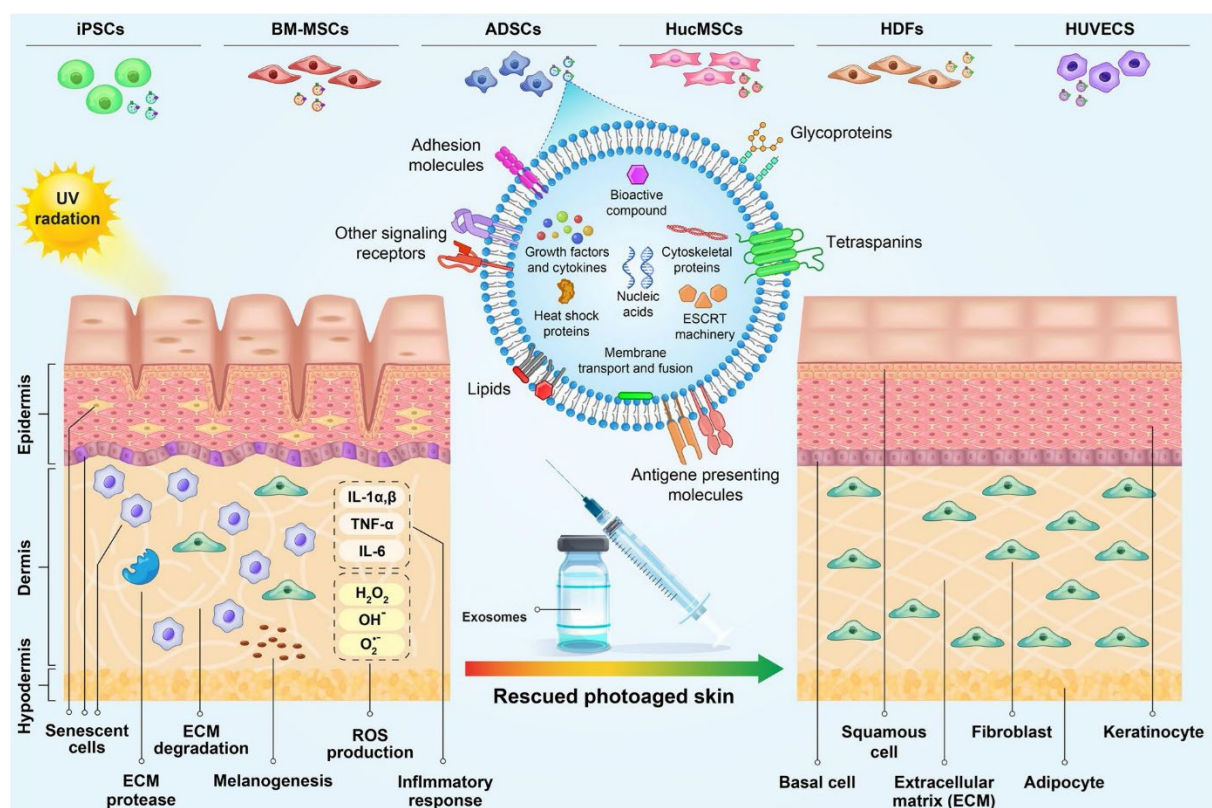
CENTELLA ASIATICA EXTRACT EXOSOMES

Exosomes are extracellular nano-sized vesicles (ranging in size from 40 to 160 nm) that transport essential biomolecules including lipids, proteins and nucleic acids (RNAs, mRNA, DNA), and metabolites required for cellular physiology as well as signalling molecules that facilitate intercellular communication. They function as messengers for tissue repair, immune system modulation, and cell-to-cell communication.

Exosomes are produced from within the endosomal compartment of cells and can be released by donor cells to modulate the function of recipient cells.

These small vesicles are secreted by most types of cells to communicate with each other. Since exosomes circulate through bodily fluids, they can transfer information not only to local cells but also to remote cells. Therefore, exosomes are considered potential biomarkers for various treatments (as they are composed of clinically important biomolecules that reflect the physiological status of the cell).

Exosomes encapsulate biomolecules within membranes that are studded with transmembrane proteins and enriched with cholesterol, sphingomyelin, and lipid rafts. These features enable exosomes to delivery internal cargo to neighbouring cells.



The unique properties of exosomes can be exploited for use as biomarkers and targeted drug delivery vehicles, and are, for this reason, gaining increasing attention in the field of dermatology.

The key benefits of exosomes are the high stability, non-immune rejection, and direct stimulation of target cells.

Exosomes have significant regenerative and cosmetic potential on account of their ability to stimulate neo angiogenesis, reverse fibroblast senescence and keratinocyte atrophy, promote synthesis of extracellular matrix components, enhance wound healing processes, and regulate oxidative stress and inflammation.

Exosomes can restore skin physiological function and regenerate or rejuvenate damaged skin tissue through various mechanisms such as decreased expression of matrix metalloproteinase (MMP), increased collagen and elastin production, and modulation of intracellular signalling pathways as well as, intercellular communication.



These molecules have several therapeutic and anti-ageing benefits. They help promote healing, hydration, elasticity, and the protection of the skin. Exosomes boost collagen production, reduce inflammation, and protect the skin from environmental stressors.

Additionally, exosomes can help to increase the efficacy of other active ingredients, such as hyaluronic acid, peptides, and antioxidants.

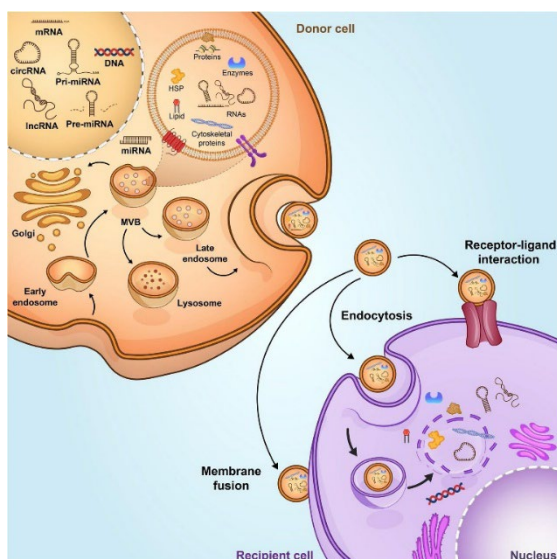
EXOSOMES BIOGENESIS

Exosomes are small membrane vesicles that are formed by internalization of plasma membrane. They are a class of microvesicular bodies (MVB), having a unique disc- or cup-shaped morphology.

After exosomes are synthesized by different cell types, they are secreted into, and transported through, various body fluids such as cerebrospinal fluid (CSF), blood, urine, and saliva.

Exosomes are generated via an endosomal pathway in which the invagination of a cell membrane and encapsulation of cytosolic components results in the formation of early endosomes.

These early endosomes undergo a maturation process through alterations in their biomolecular composition that leads to the formation of late endosomes.



During this maturation process, the endosomal membrane invaginates to form intraluminal vesicles (ILVs) in the lumen of organelles.

The microvesicular bodies then either undergo fusion with the lysosome membrane and follow a degradative pathway, or they fuse with the plasma membrane and are secreted into extracellular space.

The secreted ILVs are generally called exosomes or extracellular vesicles (EVs).

Exosomes contain various biomolecules depends on the cell type of origin. Lipids, proteins and nucleic acids are the common molecular constituents of the majority of exosomes. Exosomes are also rich in cytokines, growth factor and antioxidants.

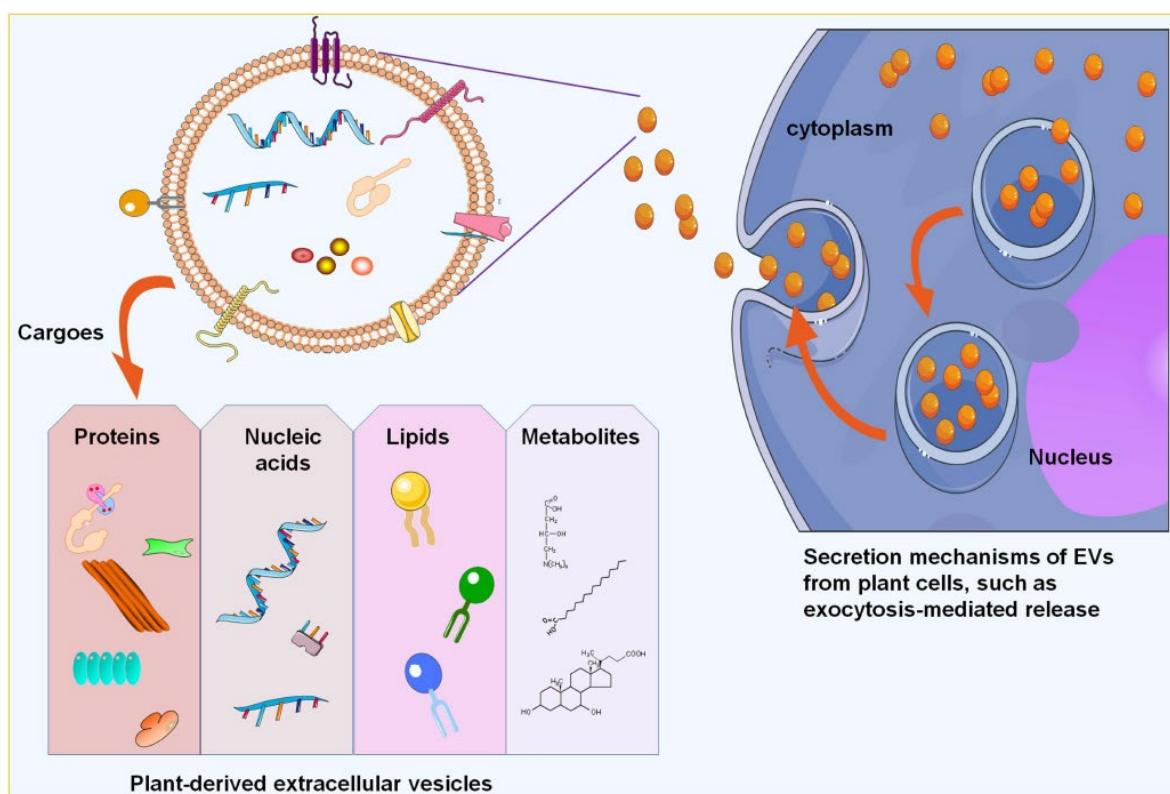
PLANT-DERIVED EXOSOMES

Regulatory and safety concerns contribute to the hesitation surrounding human-derived exosomes. Since exosomes are derived from cells, it is imperative to establish standardised manufacturing processes and quality control measures to ensure their safety, efficacy, and reproducibility.

Additionally, thorough evaluation of the potential off-target effects or unintended consequences of human exosome-based therapies is still ongoing. This is where plant-based alternatives offer an excellent opportunity.

Plant-derived exosomes are nanovesicles that contain biomolecules, which protect plants from stress and pathogens. Over the past decade, research has unveiled that plant stem cells produce human cytokine-like factors that enable the use of plant secretory factors for human cell signalling.

For aesthetic applications, this means scientists can load these vesicles with thousands of biomimetic factors, including growth factors, DNA, RNA, miRNA, peptides, liposomes, amino acids, and proteins directed to target inflammation, wound healing, angiogenesis and the stimulation of hyaluronic acid, collagen and elastin production.



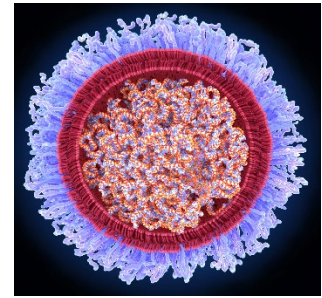
Plant-based exosomes have an innate biocompatibility inherent in plant-derived materials. This quality plays a crucial role in mitigating the tendency for immunogenic responses and toxic manifestations, commonly observed with synthetic and human analogues. Plant-based exosomes can safely and effectively interact with recipient cells without triggering detrimental immunogenic responses.

Plant-based exosomes are composed of lipid bilayers that enclose a cargo-rich interior, providing exceptional stability and protecting the encapsulated cargo from degradation. These nanoparticles comprise proteins, lipids, nucleic acids, and metabolites, being ideal for intercellular transport, and starting numerous biological processes.

Plant exosomes are composed of the same phospholipid bilayer as skin cells, so they have excellent skin penetration.

Coated with biocompatible materials such as lipids, plant exosomes offer several advantages:

- Firstly, they provide a protective layer that shields the nanoparticles from immune surveillance and minimizes their interaction with immune cells.
- This protective coating enhances the stability and biocompatibility of exosomes, allowing them to efficiently traverse physiological barriers. Specific ligands can be conjugated to the surface of the exosomes to enhance their target specificity, enabling selective interactions with desired cell types or tissues.
- In addition, they contain different surface proteins and molecules that play a crucial role in cell targeting and interaction, enhancing the efficacy in tissue engineering.



Centella asiatica have wound healing and regenerative properties, antioxidant and antibacterial activity, and immunomodulatory function. Exosomes are involved in plant growth and development, defense reactions, and symbiosis between plants and microorganisms.

Centella asiatica exosomes mimic human ones in size and shape, offering a more viable, safe and effective option. Plant-derived exosomes are known to have very high stability and biocompatibility in the body because they are not toxic or immunogenic compared with exosomes isolated from mammalian cells.

Exosomes isolated from mammalian cells contain about 20% cholesterol; however, plant-derived exosomes have been reported to have no cholesterol.

Plant-derived exosomes have the advantage of being able to be stored for a long time due to their long persistence, making them easy to store.

They have a high safety profile, excellent absorption capacity and effectiveness, having a low possibility of side effects due to their high biocompatibility.

Furthermore, they demonstrate exceptional capabilities in encapsulating and transport nucleic acids, proteins, lipids, and other therapeutic agents.

By controlling the release of bioactive molecules, plant exosomes can target specific areas and provide sustained and localized therapy (anti-inflammatory, wound healing, increase of collagen and elastin production, stimulation of blood vessel formation...).



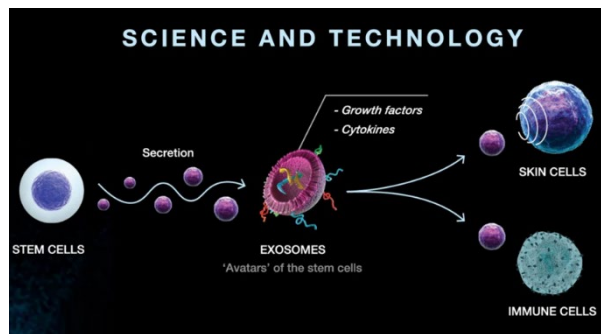
EXOSOMES BENEFITS

THE REGENERATIVE POWER OF EXOSOMES

- **REDUCTION OF PHOTO-OXIDATIVE STRESS & ANTI-INFLAMMATORY EFFECT**

Ageing skin is particularly vulnerable to the detrimental effects of oxidative stress and inflammation due to fibroblast senescence, which occurs because of ageing. Exposure of skin to UV radiation generates hydrogen peroxide (H₂O₂) and various other reactive oxygen species (ROS) which contributes to the development of skin pathologies and accelerates the ageing process.

The application of exosomes reduces ROS production and inflammation, thanks to its antioxidant signalling capacity.



Exosomes enhance cell viability, suppress ROS induced - DNA damage, and inhibit the expression of matrix metalloproteinases (MMPs).

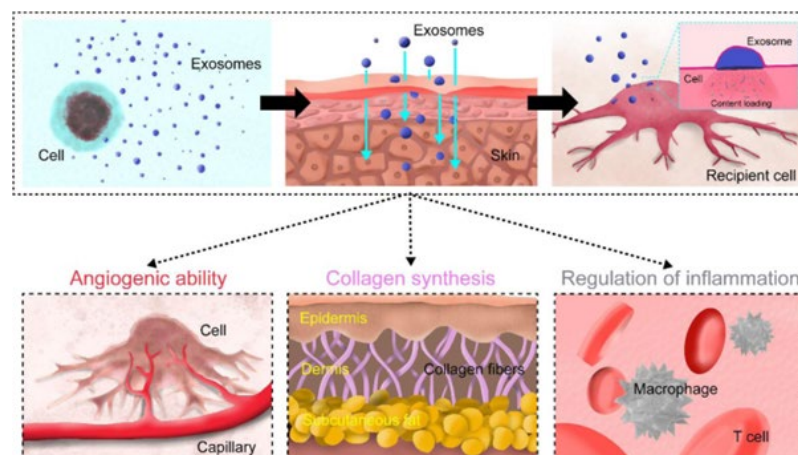
Exosomes modulate the immune system to reduce inflammation and improve skin health. The immunomodulatory properties of exosomes can help reduce inflammation in the skin, which is a significant contributor to skin aging.

- **PROMOTION OF COLLAGEN SYNTHESIS**

Exosomes promote rejuvenation and repair of photodamaged skin, by increasing collagen production by dermal fibroblasts, as well as promoting thickening of the dermal matrix. They contain growth factors and signalling molecules that can stimulate the synthesis of collagen and other extracellular matrix proteins, helping to maintain skin elasticity and firmness, and reduce wrinkles and fine lines.

Additionally, exosomal proteins and lipids plump and hydrate the skin, improving its texture. Ingredients of exosomes such as cytokines, nucleic acids, proteins, and other bioactive compounds can also help to protect the skin from environmental stressors and reduce the appearance of dark spots and other discoloration. Exosomes can help to repair skin damage such as sun damage and acne scars.

With their ability to help improve skin tone, texture, and appearance, exosomes provide several promising therapeutic and anti-aging benefits.



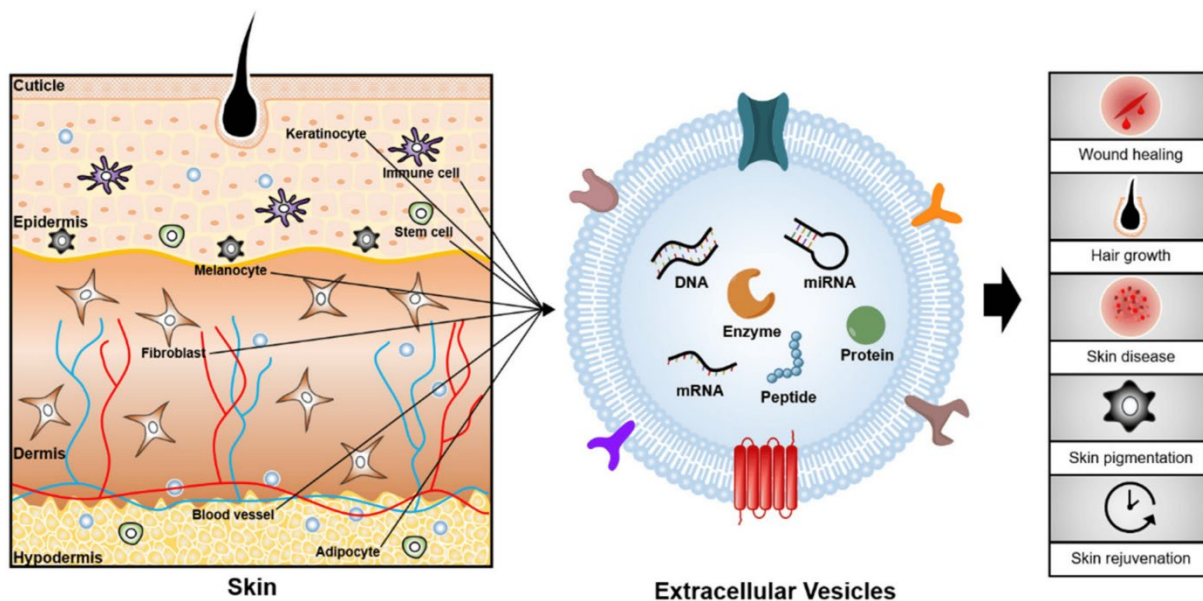
- **ACCELERATE WOUND HEALING**

Exosomes can promote wound healing by stimulating cell migration, proliferation, and differentiation, as well as by modulating the immune response and promoting angiogenesis. This property could be beneficial for the repair of skin damage caused by environmental factors or the natural aging process.

Exosomes contain pro-inflammatory proteins that stimulate the release of cytokines that initiate the recruitment of immune cells to the wound site and activate wound healing.

In addition, exosomes contain growth factors that stimulate the growth of fibroblasts, which helps to repair the wound.

Exosomes also promote cell-to-cell communication. They contain miRNAs that are proven to regulate gene expression in the cells around the wound. This promotes the formation of new blood vessels, increasing the blood flow to the wound site. This increased blood flow then delivers oxygen and nutrients to the wound, which further speeds up the healing process.



Centella asiatica extract is a rich source of natural bioactive substances, triterpenoid saponins, flavonoids, phenolic acids, triterpenic steroids, amino acids and sugars. Thus, many scavenging free radicals, exhibit anti-inflammatory activity and effect on the stratum corneum hydration and epidermal barrier function.

It is effective in the treatment of small wounds, hypertrophic wounds, burns, psoriasis...etc. It can stimulate the fibroblast proliferation and activate the SMAD signalling pathway therefore can increase the type I collagen production as well as decrease of stretch marks and inflammatory reactions.

Moreover, the constituents of *Centella asiatica* can increase the microcirculation of blood in the skin ensuring optimal oxygenation of nutrition of tissues.

With soothing and toning properties. Promotes collagen synthesis and cicatrisation. It protects the blood vessels and increases their oxygenation, thus reducing swelling and water retention.

Research Progress of *Centella asiatica*-derived Agents

FUNCTIONS	STUDIES
Anti-inflammatory	<i>Centella asiatica</i> phospholipid complexes showed significant anti-inflammatory effects in a mouse model of phthalic anhydride-induced atopic dermatitis. Hydroxy <i>Centella asiatica</i> phospholipids significantly inhibited the nuclear translocation of pro-inflammatory cytokines IL-2 β , Toll-like receptor 1, and NF- κ B in THP-1 human monocytes stimulated by <i>Propionibacterium acnes</i> and inhibited acne formation.
Promoting skin damage repair	Centella extract enhances the migratory activity of human keratinocytes in a concentration- and time-dependent manner. They also promoted wound repair by promoting fibroblast proliferation and collagen synthesis during the early stages of repair and exhibited some antimicrobial activity.
Antioxidant and anti-aging	Phenolics contained in Centella extract preparations have been associated with slowing down skin photoaging effects by scavenging ROS. Plant compositions containing Centella extract reduced oxidative stress-induced telomere shortening in fibroblasts and inhibited the accumulation of glycated substances.
Improvement skin barrier hydration	<i>Centella asiatica</i> glycosides acted to enhance skin hydration by increasing water channel protein 3 expression, promoting hyaluronic acid secretion in human dermal fibroblasts, and decreasing transepidermal water loss in a methyl nicotinate model.
Anti-scarring	In a rabbit ear scar model, oral administration of <i>Centella asiatica</i> glucoside was able to inhibit keloid formation by inhibiting the expression of collagen I and III and TGF- β and promoting the expression of Smad7 and peroxisome proliferator-activated receptor gamma. Gel preparations of Centella extracts show better efficacy.

POLYNUCLEOTIDES

Polynucleotides are a newly emerging trend in anti-ageing injections that stimulate tissue repair and encourage restoration and regeneration of the skin from within.

Polynucleotides (full name polydeoxyribonucleotides) are long chains of nucleotides that make up DNA and RNA molecules (with a molecular weight of 250-500 kDa).

They are derived from fragments of salmon DNA (which resemble human DNA), which means they have the ability to function as a biostimulator, kick-starting regeneration in the skin.



Are a type of messenger that work by signalling to particular cells in the body to change their behaviour and start processes of regeneration, repair and collagen production.

Benefits: collagen and elastin production (increase elasticity, reduce wrinkles), reduce inflammation, antioxidant, increase hydration, dermis re-densifying, reduce hyperpigmentation. DNA AND RNA segments with a three-dimensional effect have the ability to attach large amounts of water and strongly hydrate the skin.

SODIUM DNA

Sodium DNA is a polynucleotide with antioxidant, repairing, regenerating and protective activity at the cellular level.

Some segments of DNA donate Purine and Pyrimidine nucleotides that are key molecules for the vitality of the cells, providing sources of energy that drive most of the reactions. The cells use the acquired sodium DNA to synthesize the nucleic acids and to metabolize their own DNA.



Through activation of the adenosine A2 receptor on the surface of fibroblasts, indirect signal stimulation of cell function occurs. Increases the production of collagen and other components of the intercellular matrix, including hyaluronic acid.

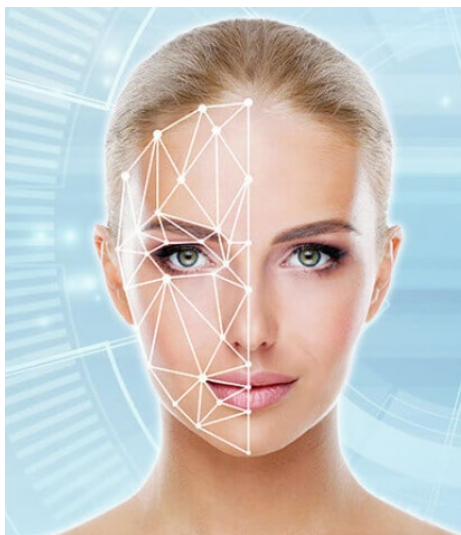
It has shown its capacity to protect and repair skin lesions caused by external agents such as UV radiations and oxidative stress, increasing wound healing processes.

Sodium DNA acts to stimulate cell repair activity, restoring damaged cells and regenerating epithelial and granulation tissues, and reduce the symptoms of inflammation, thus accelerating the healing of skin micro lesions.

It avoids collagen destruction, carrying out a protective function from damage induced by free radicals. It stimulates cell proliferation, improving the vitality of fibroblasts and the rebuilding of the connective tissues, leading to a more elastic and hydrated skin with a thicker dermis and a significant decrease of wrinkles.

Extracted from wild salmon milt from the Pacific Ocean.

HYDROLIZED DNA / RNA



DNA-RNA Complex contains hydrolysed RNA and DNA segments of low molecular weight, known as nucleotides, that fight skin dehydration.

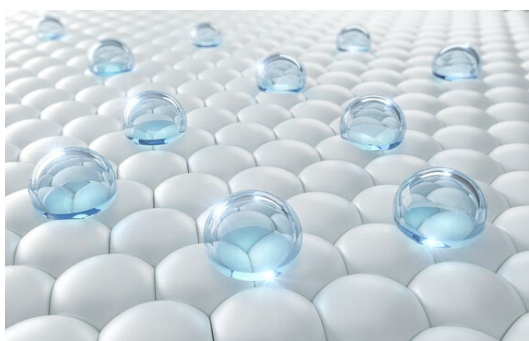
At the same time, they significantly enhance the functional activity of skin cells by producing collagen and protecting the cell membrane from free radicals and oxidative stress.

Nucleic Acids, so-called because they were first isolated from cell nuclei about 100 years ago, are of two general types: RNA and DNA. Like proteins and polysaccharides, nucleic acids are huge molecules. Each nucleic acid consists of either one or two chains of nucleotides. Each nucleotide, in turn, is composed of three covalently bonded units, a heterocyclic base (purines and pyrimidines), ribose or deoxyribose, and a phosphate.

These nucleotides and nucleosides provide skin moisturizing and tonifying effects.

Hydrolyzed DNA and Hydrolyzed RNA are made by taking DNA and RNA through a fraction (hydrolysis) process to make the molecular size smaller so that it can be absorbed into the skin. They provide structural cell reinforcement and renewal to prevent chronological ageing.

HYALURONIC ACID



Hyaluronic Acid is a natural polysaccharide found in the connective tissue throughout the body with the unique ability to retain 1000 times its weight in water, providing the cells with a deep hydrating and lubricating activity.

It attracts and binds water and fills the space between collagen and elastin fibers, helping to increase skin hydration and maintain its volume and turgidity. Hyaluronic acid acts as a cushion, free radicals capturer and lubricant to reduce skin damage, being an essential component of youthful, radiant, supple and elastic skin.

It promotes the transport of essential nutrients to skin cells, maintains its normal metabolism and acts as a protecting agent against external damage, due to its free radical scavenging properties.

Hyaluronic acid participates in the processes of skin repair and contributes to dermis renewal by activating fibroblasts activity, restructuring collagen fibers, enhancing the synthesis of glycosaminoglycans and growth factors. It provides structural support to the skin, increasing its elasticity while eliminating the wrinkles.

Hyaluronic acid allows a better tissue vascularization and cell nutrition enhancing the proliferation of endothelial cells and the increase of cutaneous brightness and glow.

ROSA CENTIFOLIA LEAF CELL EXTRACT

Plant cells stimulate and compensate for the functions of epidermal stem cells.

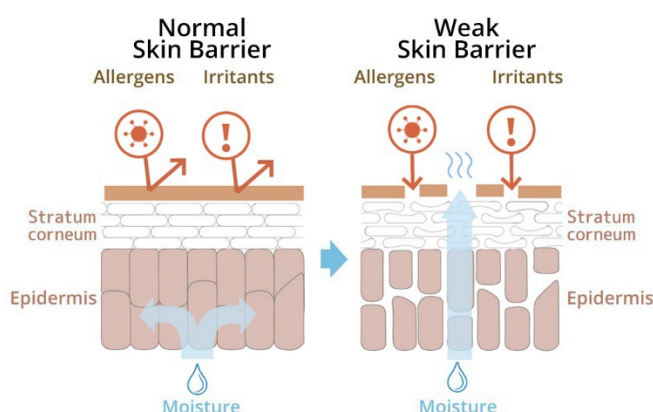
Also known as mother cells, these cells are in perpetual renewal. Indeed, their role is to preserve the health of the skin and protect the epidermal tissue against external aggressions. However, with age and under the action of various factors, the function of stem cells slows down, leading to skin ageing, dehydration, etc.

To counteract this decrease in activity, the use of external stem cells should be considered.

The antioxidant properties of the Rosa centifolia particularly help to combat the formation of unstable oxygen molecules responsible for skin ageing.

As a moisturising agent, these plant cells also replenish the skin layer with water. This plant extract restores the content of moisturising agents and lipids naturally present in the horny layer to replenish the hydrolipidic film of the skin. This allows the epidermal tissue to regain volume. The skin, thus plumped, becomes more radiant.

They keep water in the epidermis, stimulates NMF levels and the quality of the hydrolipid film in the corneus layer. By increasing epidermal cell regeneration, it helps to rebuild the hydrolipidic film, reinforcing the stratum corneum.



In addition, they contribute to the production of lipids (ceramides, free fatty acids, cholesterol...), main elements of the stratum corneum of the epidermis, and help to keep corneocytes together.

This way they improve hydrolipidic balance, reducing water loss and providing a better protection against external aggressors.

It is important to note that plant native cells are a result of the totipotency of plant cells (reconstruction with DNA). The advantage of this biotechnological process lies in the targeted selection of stem cells and the control of the culture environment.

This is carried out in a sterile environment, that is, one free from microorganisms. Plant native cells retain all the natural elements of the plant and have a molecular richness (amino acids, vitamins, minerals).

The native cells of *Rosa centifolia* help to attract and retain water in the epidermal tissue. They also maintain the skin's radiance and strengthen the skin barrier against external aggressions.

The plant cells of the hundred-petaled rose (rich in vitamins, minerals and antioxidants) are therefore hydrating, antioxidant, and soothing active ingredients. They calm skin, reducing redness and inflammation.

It stimulates regenerative processes and skin cell proliferation, activating fibroblasts to produce collagen and elastin, key proteins responsible for skin elasticity and firmness. Visibly reduce signs of aging.



CENTAUREA CYANUS CALLUS LYSATE EXTRACT

Skin cells are damaged, specifically because of their exposure to free radicals that modify their structure, causing them to function less efficiently and ultimately affecting their DNA. This provokes premature ageing. Centaurea cyanus callus lysate extract can slow down this process and stimulate the repair of cell functions, especially the fibroblasts to keep a firmer skin with less wrinkles.

Rich in polyphenols (flavonoids, tannins and anthocyanin) and with anti-inflammatory and antioxidant properties, Centaurea Cyanus Callus Lysate Extract protects the skin from external influences and free radical damage.



It creates a protective barrier on the skin to defend it from harmful substances, irritants, allergens, pathogens that can cause various inflammatory conditions.

Centaurea Cyanus Callus Lysate Extract has a natural lifting, anti-ageing, and anti-wrinkle effect. It promotes the proliferation of fibroblasts and keratinocytes, key cells which support tissues in the body.

It stimulates synthesis of type 1 and type 3 collagen, elastin and polysaccharides, essential for toned, smooth skin.

It improves fibroblast function by protecting them from physiological oxidative stress and by stimulating their multiplication.

Centaurea Cyanus Callus Lysate Extract restores the synthesis of components essential to the skin's firmness: collagens fibres and polysaccharides. In addition, it relaunches the proliferation of keratinocytes, essential for the skin's density and surface hydration.

PANTHENOL

Panthenol acts as an effective moisturizer, reducing trans-epidermal water loss, improving stratum corneum hydration and maintaining skin softness and elasticity. It nourishes, protects and regenerates the scalp, restoring hair natural glow.

Moreover, it activates the fibroblast proliferation and accelerates cell regeneration and the re-epithelisation in wound healing, helping repair damaged tissues.

Dexpanthenol is essential for the normal function of epithelial tissues, promoting normal keratinization of the skin and hair, being also an effective calming agent.

Panthenol improves the health of the adrenal glands, impeding hormonal imbalances that may increase DHT levels, avoiding the outbreak of acne.

Panthenol plays a central role in lipid metabolism. Coenzyme A, involved in more than 100 reactions, is the most important coenzyme in the body. It is formed from adenosine triphosphate, cysteine, and Pantothenic acid. It is essential in fatty acid metabolism and sex hormone synthesis.

With a shortage of Panthenol, Coenzyme A cannot perform its functions property. Fatty acid metabolism is for the time being at least in part halted, lipid in the sebaceous glands starts to accumulate, sebum excretion is increased, and acne begins to appear. Panthenol increases the amount of Coenzyme A, which increases the metabolic breakdown of oils by normal activity of cell physiology.

It decreases sebum secretion, becoming the skin less oily. It regresses existing acne lesions and slows down the eruption of new ones, reducing the pore size and leading to a much finer, smoother, and more beautiful skin.

Cell^{Switch} technology for skin regeneration and rejuvenation

